

| Type                                  | Interval (s) | CAN Device Identifier (11-bit) |                            | Data (8-byte)                                                                                                                                           |                                       |
|---------------------------------------|--------------|--------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|                                       |              | Device Identifier (6-bit)      | Message Identifier (5-bit) | Lower Datum (4-byte)                                                                                                                                    | Higher Datum (4-byte)                 |
| Cell voltages                         | 1            | 48                             | 0                          | Cell 0.0 voltage (float)                                                                                                                                | Cell 0.1 voltage (float)              |
|                                       |              |                                | 1                          | Cell 0.2 voltage (float)                                                                                                                                | Cell 0.3 voltage (float)              |
|                                       |              |                                | 2                          | Cell 0.4 voltage (float)                                                                                                                                | Cell 0.5 voltage (float)              |
|                                       |              |                                | 3                          | Cell 1.0 voltage (float)                                                                                                                                | Cell 1.1 voltage (float)              |
|                                       |              |                                | 4                          | Cell 1.2 voltage (float)                                                                                                                                | Cell 1.3 voltage (float)              |
|                                       |              |                                | 5                          | Cell 1.4 voltage (float)                                                                                                                                | Cell 1.5 voltage (float)              |
|                                       |              |                                | 6                          | Cell 2.0 voltage (float)                                                                                                                                | Cell 2.1 voltage (float)              |
|                                       |              |                                | 7                          | Cell 2.2 voltage (float)                                                                                                                                | Cell 2.3 voltage (float)              |
|                                       |              |                                | 8                          | Cell 2.4 voltage (float)                                                                                                                                | Cell 2.5 voltage (float)              |
|                                       |              |                                | 9                          | Cell 3.0 voltage (float)                                                                                                                                | Cell 3.1 voltage (float)              |
|                                       |              |                                | 10                         | Cell 3.2 voltage (float)                                                                                                                                | Cell 3.3 voltage (float)              |
|                                       |              |                                | 11                         | Cell 3.4 voltage (float)                                                                                                                                | Cell 3.5 voltage (float)              |
|                                       |              |                                | 12                         | Cell 4.0 voltage (float)                                                                                                                                | Cell 4.1 voltage (float)              |
|                                       |              |                                | 13                         | Cell 4.2 voltage (float)                                                                                                                                | Cell 4.3 voltage (float)              |
|                                       |              |                                | 14                         | Cell 4.4 voltage (float)                                                                                                                                | Cell 4.5 voltage (float)              |
|                                       |              |                                | 15                         | Cell 5.0 voltage (float)                                                                                                                                | Cell 5.1 voltage (float)              |
|                                       |              |                                | 16                         | Cell 5.2 voltage (float)                                                                                                                                | Cell 5.3 voltage (float)              |
| Thermistor temperatures               | 1            | 48                             | 17                         | Cell 5.4 voltage (float)                                                                                                                                | Cell 5.5 voltage (float)              |
|                                       |              |                                | 18                         | Thermistor 0.0 temperature (float)                                                                                                                      | Thermistor 0.1 temperature (float)    |
|                                       |              |                                | 19                         | Thermistor 0.2 temperature (float)                                                                                                                      | Thermistor 1.0 temperature (float)    |
|                                       |              |                                | 20                         | Thermistor 1.1 temperature (float)                                                                                                                      | Thermistor 1.2 temperature (float)    |
|                                       |              |                                | 21                         | Thermistor 2.0 temperature (float)                                                                                                                      | Thermistor 2.1 temperature (float)    |
|                                       |              |                                | 22                         | Thermistor 2.2 temperature (float)                                                                                                                      | Thermistor 3.0 temperature (float)    |
|                                       |              |                                | 23                         | Thermistor 3.1 temperature (float)                                                                                                                      | Thermistor 3.2 temperature (float)    |
|                                       |              |                                | 24                         | Thermistor 4.0 temperature (float)                                                                                                                      | Thermistor 4.1 temperature (float)    |
|                                       |              |                                | 25                         | Thermistor 4.2 temperature (float)                                                                                                                      | Thermistor 5.0 temperature (float)    |
|                                       |              |                                | 26                         | Thermistor 5.1 temperature (float)                                                                                                                      | Thermistor 5.2 temperature (float)    |
| High Voltage Bus voltage and current  | 0.1          | 48                             | 27                         | HV Bus voltage (float)                                                                                                                                  | HV Current (float)                    |
| Statuses                              |              |                                | 28                         | [0: relay status][1: electric safe discharge status][2: discharge status][3-23: reserved][24-31: flag_hold]                                             | Supplementary battery voltage (float) |
| Overvoltage/temperature/current flags |              |                                | 29                         | [0-35: Cell 0.0, 0.1, ..., 5.5 overvoltage flags][36-53: Thermistor 0.0, 0.1, ..., 5.2 overtemperature flags][54: overcurrent flag][55-63: reserved]    |                                       |
| Undervoltage/temperature flags        | 1            | 48                             | 30                         | [0-35: Cell 0.0, 0.1, ..., 5.5 undervoltage flags][36-53: Thermistor 0.0, 0.1, ..., 5.2 undertemperature flags][54: undercurrent flag][55-63: reserved] |                                       |
| Low Voltage Bus voltage and current   |              |                                | 31                         | LV Bus voltage (float)                                                                                                                                  | LV Current (float)                    |